

Lab Report 05: RIP Routing Protocol

Experiment Name: Configuration and Verification of RIP Routing in a Multi-Router Topology

1. Introduction

The **Routing Information Protocol (RIP)** is a dynamic distance-vector routing protocol that uses **hop count** as its primary metric. RIP prevents routing loops by implementing a limit on the number of hops allowed in a path from source to destination (maximum of 15). In this experiment, we implement **RIP Version 2**, which supports Classless Inter-Domain Routing (CIDR) and provides more efficient routing updates compared to Version 1.

2. Tools

- **Cisco Packet Tracer** (Simulation Environment)
 - **3x Routers** (Router-PT)
 - **3x Switches** (2960 Series)
 - **6x PCs** (End Devices)
 - **Cables:** Copper Straight-Through and Serial DCE.
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3. Procedure

Step 1: Physical Topology Setup

The network was designed with three routers connected in a chain via serial interfaces. Each router is connected to a local switch, which in turn connects two PCs.

- **Router0** handles the 10.10.10.0subnet.
- **Router1** handles the 20.20.20.0subnet.
- **Router2** handles the 30.30.30.0subnet.

Step 2: IP Configuration

IP addresses were assigned to the FastEthernet and Serial interfaces. Default gateways were set on all PCs to match their respective router interface IPs.

Step 3: RIP Configuration

To enable dynamic routing, the following commands were executed on each router (example for Router1):

Bash

```
Router1> enable
```

```
Router1# configure terminal
```

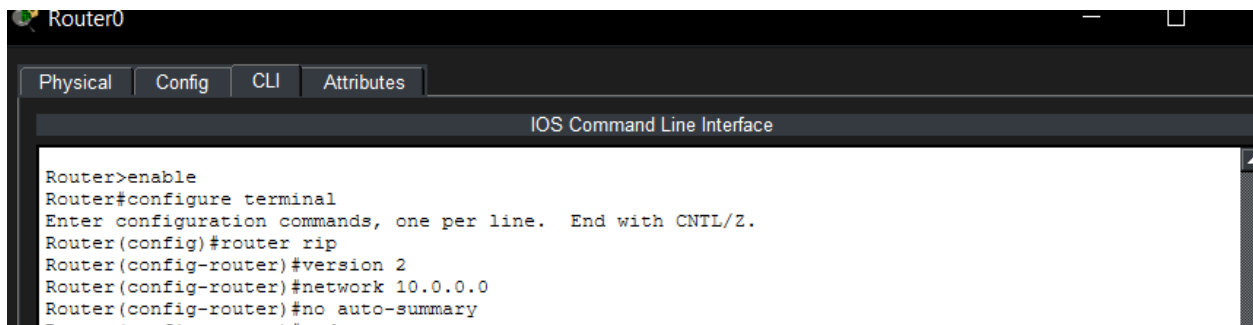
```
Router1(config)# router rip
```

```
Router1(config-router)# version 2
```

```
Router1(config-router)# network 20.0.0.0
```

```
Router1(config-router)# network 10.0.0.0 -- (Example for Serial Link Network)
```

```
Router1(config-router)# no auto-summary
```

A screenshot of a network simulator window titled "Router0". The window has a dark background with a light-colored terminal area. At the top, there are tabs labeled "Physical", "Config", "CLI", and "Attributes", with "CLI" being the active tab. Below the tabs, the text "IOS Command Line Interface" is displayed. The terminal area shows the following commands and prompts:

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
```

Figure 1: RIP Configuration Commands in CLI

4. Verification

A. Routing Table Check

To verify that RIP is working, the show ip route command was used. The presence of the letter "R" next to a route indicates it was learned successfully via the RIP protocol.

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#network 10.0.0.0
Router(config-router)#no auto-summary
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C 10.0.0.0/8 is directly connected, FastEthernet0/0
R 20.0.0.0/8 [120/1] via 40.40.40.2, 00:00:16, Serial2/0
R 30.0.0.0/8 [120/2] via 40.40.40.2, 00:00:16, Serial2/0
C 40.0.0.0/8 is directly connected, Serial2/0
R 50.0.0.0/8 [120/1] via 40.40.40.2, 00:00:16, Serial2/0

Router#
Copy Paste
Top

```

Figure 2: Routing Table showing dynamic routes (marked with R)

B. Connectivity Test

Using the PDU (Ping) tool in Packet Tracer, packets were sent between PCs on different subnets (eg, PC0 to PC5).

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC2	PC3	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC4	PC5	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	3	(edit)	(delete)

Figure 3: Successful ICMP Connectivity Results

5. Conclusion

In this lab, we successfully configured a three-router topology using the **RIP Routing Protocol** . The experiment demonstrated that RIP can automatically calculate and populate routing tables, allowing devices on the 10.10.10.0network to communicate with the 30.30.30.0network without static route entries. The successful ping results and the "R" entries in the routing table confirm that the protocol is functioning correctly as a dynamic gateway for small-scale networks.